

SIMATS ENGINEERING

Short Answer Test

COURSE NAME: Software Testing

COURSE CODE: CSA3711

COURSE OUTCOME COVERED

CO1: *Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.*

Assessment Tool Weightage: 30%

Sl. No	Questions	Bloom's Level
1	A startup team relies on random testing methods, leading to inconsistent results and missed defects. The project manager wants to improve testing effectiveness. As a test engineer, apply the concept of testing as an engineering activity to redesign the approach. Explain how structured test design improves quality and defect detection. Answer: Testing should be treated as a systematic engineering activity instead of random testing. The team should first understand the requirements, prepare a test plan and design test cases with clear inputs and expected results. Techniques such as equivalence partitioning and boundary value analysis can be used. Structured test design provides better coverage, makes testing repeatable and helps identify defects that may be missed during random testing.	3
2	A banking system undergoes multiple failures due to lack of a defined testing process. The organization plans to standardize its testing activities. Illustrate how a well-defined testing process improves software quality in such real-time systems. Explain the benefits with relevant examples. Answer: A well-defined testing process includes requirement analysis, test planning, test case design, execution, defect reporting, retesting and regression testing. For a banking system, transactions can be tested for valid transfers, insufficient balance and invalid account details. Following the	3

	same process for every release reduces missed scenarios and repeated defects. It improves consistency, reliability and overall software quality.	
3	During development of an e-commerce platform, testing activities are performed inconsistently across phases. This leads to integration and system-level defects. Demonstrate how testing should be carried out as a structured process in SDLC. Explain how process discipline ensures better quality outcomes. Answer: Testing should be performed throughout the SDLC. Testers should review requirements, prepare test scenarios during design, perform integration testing between modules and conduct system testing on the complete application. After fixing defects, retesting and regression testing should be performed. Following a common process improves communication, prevents missed defects and ensures better quality at every stage.	3
4	A login system shows mismatch between expected and actual outputs during validation. The development team is unsure whether the issue is an error, defect, or failure. Apply basic testing definitions to classify the issue. Explain the relationship between these terms with suitable examples. Answer: An error is a human mistake, such as a developer writing an incorrect password condition. A defect is the incorrect code introduced because of that error. A failure occurs when the software executes the defective code and produces an incorrect result, such as allowing an invalid password. Therefore, the relationship is Error → Defect → Failure.	3
5	A team struggles to detect defects early, leading to increased cost during later stages. The manager wants to improve defect detection efficiency. Apply a software testing principle to address this issue. Explain how the principle improves testing effectiveness with an example. Answer:	3

	The principle of early testing should be applied. Testers should participate from the requirement and design stages instead of waiting until coding is completed. For example, an unclear banking transaction requirement can be identified during requirement review itself. Early defect detection reduces rework, saves cost and prevents defects from moving to later stages.	
6	During requirement analysis of a healthcare system, several ambiguities go unnoticed. Later, these lead to defects in implementation. Show how a tester can contribute during the requirement phase to improve quality. Explain the importance of early testing involvement. Answer: A tester can review requirements and identify ambiguous, incomplete or missing conditions during the requirement phase. The tester can also check whether each requirement is clear and testable. For example, if a healthcare requirement says patient data must be secure, the tester can ask for specific security conditions. Early involvement prevents misunderstandings, reduces rework and improves software quality.	3
7	A payroll system produces incorrect outputs due to unclear and ambiguous requirements. The development team needs to identify the origin of defects. Analyze how such defects originate in the software lifecycle. Explain how requirement clarity can prevent such issues. Answer: Defects can originate when unclear requirements are misunderstood by developers during design and coding. For example, if overtime salary calculation rules are not clearly defined, different developers may implement them differently. Requirement reviews, discussions with stakeholders and clear acceptance criteria can prevent such problems. Clear requirements provide a common understanding and reduce defects in later stages.	3
8	A system crashes when boundary input values are entered by users. The team suspects improper testing coverage. Classify this defect and justify your answer. Explain how boundary value analysis could have prevented this issue. Answer:	3

	This can be classified as a functional or validation defect because the system does not correctly handle boundary inputs. Boundary Value Analysis tests values at and around the limits. For example, for an allowed range of 1 to 100, values such as 0, 1, 2, 99, 100 and 101 should be tested. This could have detected the defect before deployment.	
9	A company maintains a defect repository but rarely uses it for improving testing practices. This results in repeated defects across releases. Demonstrate how a defect repository can support better test case design. Explain its role in improving software quality. Answer: A defect repository stores information about previous defects, their causes, affected modules and resolutions. Testers can analyze this data to identify repeated defect patterns and create additional test cases for those areas. For example, if many defects occur in the payment module, more payment-related test cases can be included in future testing. Thus, defect history improves test coverage and helps prevent recurring defects.	3
10	An ATM system fails during withdrawal due to incorrect handling of transaction logic. The issue was not detected during testing. Apply your testing knowledge to identify and describe this as a functional defect. Explain how it could have been prevented. Answer: This is a functional defect because the ATM is not performing the withdrawal operation according to the requirement. For example, if a user withdraws ₹2,000, the system should dispense ₹2,000 and correctly update the account balance. Proper test cases for valid withdrawal, insufficient balance, invalid amounts and transaction limits could have detected the defect. Regression testing should also be performed after fixing it.	3
11	A large application shows that most defects are concentrated in a few modules. The testing team wants to optimize their efforts. Use the concept of defect clustering to identify testing priorities. Explain how this improves testing efficiency. Answer:	3

	Defect clustering means that a small number of modules may contain most of the defects. The testing team should analyze previous defect data and give higher priority to these high-risk modules. For example, if most defects occur in the payment module, it should receive more testing attention. This helps use testing time and resources effectively and improves defect detection.	
12	A web application experiences slow performance under heavy user load. The issue needs to be categorized properly. Apply defect classification techniques to identify this type of defect. Explain how classification helps in better resolution. Answer: The issue can be classified as a performance defect because the application becomes slow under heavy user load. Load testing and stress testing can be used to identify the problem. For example, if the application works normally with 100 users but becomes slow with 5,000 users, the cause may be server, database or code performance. Correct classification helps developers focus on the appropriate solution.	3
13	A team is preparing for regression testing after multiple releases. They have access to historical defect data. Show how this data can be reused effectively for regression testing. Explain how it improves test coverage and efficiency. Answer: Historical defect data can be used to identify previously failed test scenarios and include them in the regression test suite. For example, if previous releases had login and payment defects, those test cases should be executed again after new changes. High-risk areas can also be prioritized. This improves coverage because old problem areas are not forgotten and increases efficiency by focusing on important tests.	3
14	During integration testing, multiple defects are found due to poor communication between developers and testers. The organization wants to improve collaboration. Identify the role of a tester in minimizing such defects. Explain how early involvement helps. Answer:	3

	Testers should communicate requirements, identify integration conditions and prepare test scenarios early in development. They should work closely with developers to understand how modules interact. For example, during payment and order integration, testers can verify whether payment status is correctly transferred to the order module. Early involvement reduces misunderstandings, improves collaboration and helps detect integration defects earlier.	
15	A software product repeatedly faces similar defects in every release cycle. The team needs a long-term solution. Apply defect prevention techniques to address recurring issues. Explain how prevention improves overall software quality. Answer: The team should perform root cause analysis to understand why the same defects are repeatedly introduced. Requirement reviews, code reviews, coding standards, developer training and automated testing can help prevent them. For example, if input validation defects repeatedly occur, common validation rules can be introduced. Defect prevention reduces repeated failures, saves time and improves overall software quality.	3
16	A mobile application frequently crashes due to inconsistent testing approaches. The testing team wants to standardize practices. Apply testing principles to redesign the testing strategy. Explain how structured approaches improve reliability. Answer: The team should prepare a standard testing process with common test plans, test case formats, execution procedures and defect reporting methods. Testing should cover different devices, operating systems, network conditions, normal inputs and edge cases. Functional, integration and regression testing should be performed systematically. Standardization improves test coverage and consistency and helps identify defects causing application crashes.	3
17	A company confuses debugging with testing, leading to delays in defect resolution. The team needs clarity on both concepts. Differentiate testing and debugging using a real-time scenario. Explain their roles in improving software quality. Answer:	3

	Testing is performed to identify defects or failures, while debugging is performed to find the cause of a defect and fix it. For example, if a banking application shows an incorrect balance, the tester reports the problem with expected and actual results. The developer then debugs the code to find the calculation or database error. After fixing it, the tester performs retesting and regression testing. Both activities are necessary for software quality.	
18	An online shopping system behaves differently across browsers. This leads to customer dissatisfaction. Apply testing principles to design a cross-browser testing strategy. Explain how such issues can be effectively handled. Answer: The application should be tested on all supported browsers such as Chrome, Edge, Firefox and Safari. Important functions such as login, product search, cart and payment should be tested on each browser and supported operating system. If a function fails in one browser, the browser version and steps should be recorded and the defect should be fixed and retested. Automation can also improve browser coverage and reliability.	3
19	A logistics application produces inconsistent outputs due to improper validation. The issue is not detected early. Apply testing concepts to identify gaps in validation. Explain how proper testing improves system reliability. Answer: The tester should check valid, invalid, boundary, empty and unexpected inputs to identify validation gaps. For example, if delivery quantity should be between 1 and 100, values such as 0, 1, 100 and 101 should be tested. Equivalence partitioning and boundary value analysis can improve validation coverage. Proper validation testing prevents incorrect data from entering the system and improves reliability.	3
20	A team skips structured test case design and directly executes random tests. This leads to poor defect detection. Apply testing as an engineering activity to improve the process. Explain the importance of systematic test design. Answer:	3

	Testing should be planned and performed systematically instead of randomly. Requirements should be analyzed first, followed by identification of test conditions and preparation of test cases with inputs and expected results. Techniques such as equivalence partitioning and boundary value analysis can be used. Systematic test design improves coverage, makes testing repeatable and increases the possibility of detecting important defects.	
21	A healthcare system shows data inconsistency between modules due to integration issues. The testing team needs to analyze the problem. Apply testing principles to identify root causes. Explain how proper testing ensures data consistency. Answer: Data inconsistency may be caused by integration errors, incorrect database updates, synchronization problems or improper data mapping. Testers should verify whether data is correctly transferred between related modules. For example, if updated patient details appear correctly in one module but not another, integration testing should be performed. Database validation and integration test cases help ensure that the same data is consistently maintained across modules.	3
22	A project experiences delays because testers are not involved early in development. The organization wants to improve collaboration. Explain the role of testers across SDLC phases. Show how early involvement improves quality. Answer: Testers should participate throughout the SDLC. They review requirements, identify test conditions during design, prepare test cases during development, perform integration and system testing, and conduct regression testing after fixes. Early involvement helps identify unclear requirements before they become coding defects. It improves communication between developers and testers, reduces rework and helps deliver better-quality software.	3
23	A system fails during peak usage due to untested scenarios. The testing team missed critical cases. Apply testing principles to identify gaps. Explain how better test coverage prevents such failures.	3

	Answer: The testing team should identify expected user loads and include normal, peak and higher-than-expected load conditions in testing. Load testing, stress testing and performance testing should be performed. For example, if thousands of users may access the system simultaneously, that condition should be simulated before deployment. Better test coverage identifies performance bottlenecks early and prevents failures during peak usage.	
24	A software team lacks proper defect documentation, making tracking difficult. The organization wants to improve defect management. Apply defect repository concepts to solve this issue. Explain how documentation enhances tracking and analysis. Answer: A centralized defect repository should be used to record every defect with its ID, description, affected module, reproduction steps, expected result, actual result, severity, priority and status. This makes it easier for developers to understand and fix defects and for testers to perform retesting. Historical defect data can also be analyzed to identify repeated problems and high-risk modules. Proper documentation improves tracking, communication and defect analysis.	3
25	A banking system produces repeated transaction failures due to improper testing practices. The team needs to improve quality. Apply software testing principles to redesign the testing approach. Explain how structured testing enhances reliability. Answer: The testing process should begin with requirement analysis and systematic test case design. Test cases should cover successful transactions, insufficient balance, invalid account details, duplicate transactions, transaction limits and network failures. Equivalence partitioning, boundary value analysis and regression testing should be used. Defects should be properly recorded and analyzed. Structured testing reduces missed scenarios and repeated failures and improves the reliability of the banking system.	3
26	A system shows failures due to missing edge case testing. The testers focused only on normal inputs. Apply testing principles to include edge cases. Explain how boundary testing improves	3

	defect detection. Answer: Testers should include edge cases and boundary values instead of testing only normal inputs. Boundary Value Analysis can be used to test minimum, maximum, just below and just above the allowed limits. For example, for a range of 1 to 100, values 0, 1, 2, 99, 100 and 101 should be tested. This helps identify validation and logical defects near the limits and improves defect detection.	
27	A development team struggles to maintain consistency in testing across modules. This leads to uneven quality. Apply process-based testing approaches to address this issue. Explain how standardization improves outcomes. Answer: A standardized testing process should be followed for all modules, including requirement analysis, test planning, test case design, execution, defect reporting, retesting and regression testing. Common templates and reporting formats should be used by all testers. Regular reviews can ensure that the process is followed correctly. Standardization reduces differences between testers, improves coverage and produces consistent quality across modules.	3
28	A web application fails due to improper handling of unexpected inputs. The testing team missed validation scenarios. Apply testing concepts to improve input validation. Explain how proper testing ensures robustness. Answer: Input validation should be tested using valid, invalid, empty, boundary, special-character and unexpected inputs. For example, if a form expects a numeric age, letters, negative numbers, empty values and very large values should also be tested. The application should display a proper validation message instead of crashing. Equivalence partitioning and boundary value analysis can improve coverage. Proper input validation makes the application more robust and reliable.	3

29	A software company faces frequent defect leakage into production. The management wants to reduce such issues. Apply testing principles to analyze the root cause. Explain how improved testing practices prevent leakage. Answer: Defect leakage may occur because of incomplete test coverage, inadequate regression testing, unclear requirements or insufficient testing of high-risk areas. The team should analyze production defects and improve testing in those areas. For example, if payment defects frequently reach production, payment scenarios should receive higher priority and be included in regression testing. Early testing, proper test design and defect analysis can reduce production defects and improve software quality.	3
30	A project team is unable to identify defects early due to poor testing strategies. This increases overall cost and effort. Apply testing principles to redesign the strategy. Explain how early defect detection improves efficiency. Answer: The team should apply the principle of early testing by involving testers in requirement and design reviews. Requirements should be checked for clarity before development, and systematic test cases should be prepared before execution. Unit, integration, system and regression testing should then be performed at the appropriate stages. For example, identifying an unclear requirement during review is cheaper than fixing it after deployment. Early defect detection reduces rework, cost and effort and improves overall testing efficiency.	3



Code of Conduct:

*I **Mugesh S (192411055)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *S. Mugesh.*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand